



Promoting STEAM Identity and Computational Thinking for Neurodiverse Youth Through Inclusive Game Design

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Introduction

Background:

- Individuals with disabilities make up only 3% of the STEAM workforce (NCSES, 2023).
- Neurodiverse youth (autism, ADHD, dyslexia) face challenges with cognition, executive function, and social interactions that make traditional STEAM pathways difficult (Astle et al., 2022; Friedensen et al., 2021).
- While awareness of disabilities in STEAM has grown, support for neurodevelopmental and learning differences remains limited and insufficient (Basham et al., 2020).

Gaming for Good (G4G) Project:

- G4G employs an inclusive, game-based learning approach.
- Departing from programming-heavy models that demand high executive functioning, G4G uses Nintendo Switch with Game Builder Garage to provide creative, hands-on design experiences.
- Promotes computational thinking and fosters positive STEAM identity for neurodiverse youth.

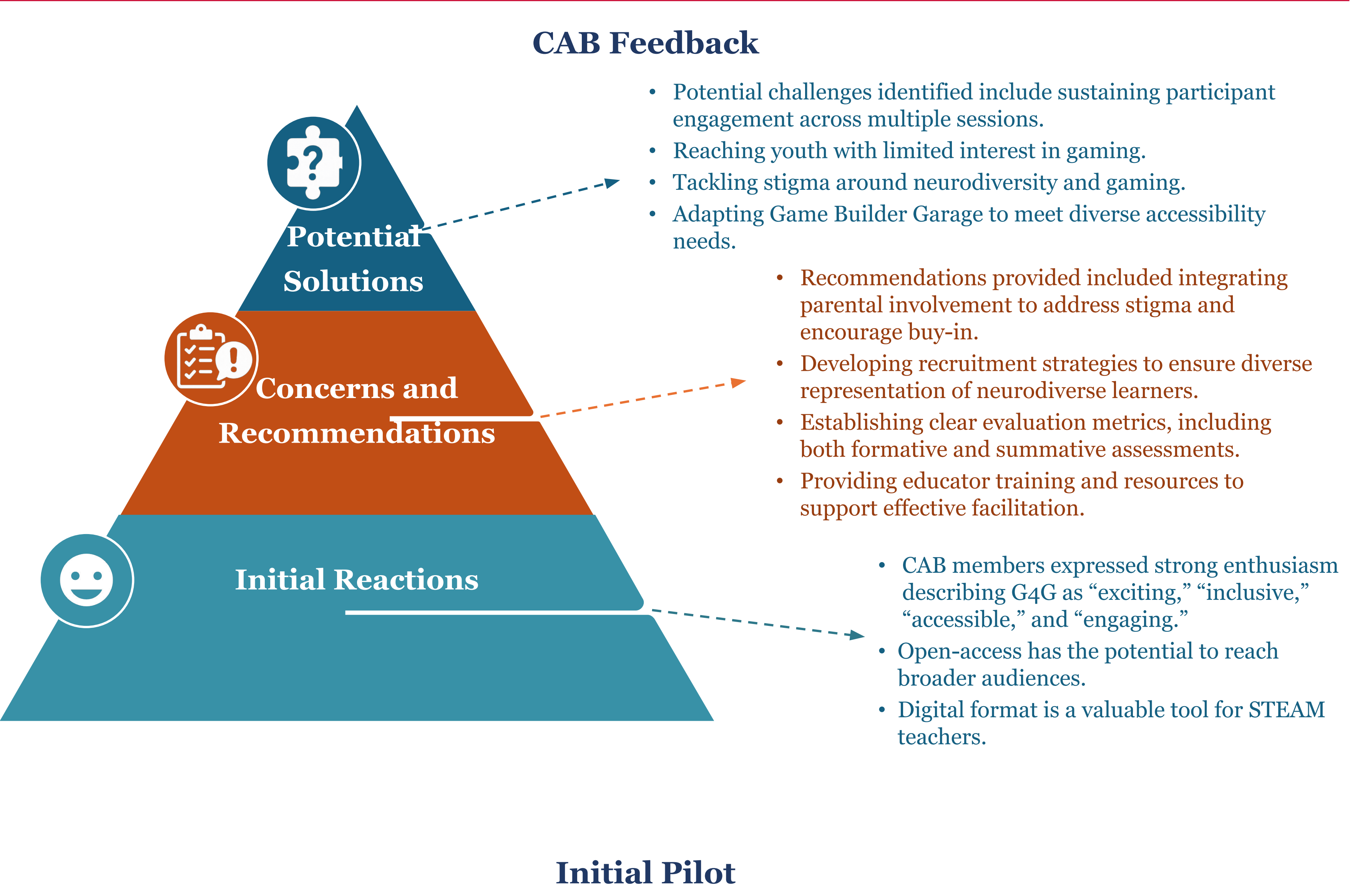
Objective:

- This study highlights our methodology, findings from our initial community engagement studios, and findings from initial pilots.

Methods

	CAB	Pilot
Study design	LXD with CES refinement, learner centered, theory informed, socio culturally responsive	Multiple case, multi method, 8 module G4G using Nintendo GBG, sites in Georgia and Missouri (schools, libraries)
Participants	n=8 (educators, informal learning experts, STEAM educators, community reps)	Enrolled n=47, analytic n=33, grades 6 to 8, neurodiverse
Data collection	Two 1 hour CES sessions on Zoom, recorded, transcribed	Field notes, classroom audio, learner artifacts, worksheets, informal interviews, site logbook, CTSS adapted for accessibility (simplified items, emoji responses)
Data analysis	Thematic coding with iterative open coding	Content analysis of qualitative data, multiple regression predicting CTSS post test from pre test and demographics

Results



Learners engaged in computational thinking and problem solving

- **CT emerged** through playful exploration, creative experimentation, and nonlinear engagement with Game Builder Garage.
- All learners **built a playable game**.
- **CT practices were evident** in decomposition, iterative refinement, and debugging.
- **Identity and self-expression** co-occurred with CT.
- **Multimodal participation extended** beyond code to original music, pixel art, and narrative design.
- **Self-efficacy** signals were frequent.
 - “I figured out I’m able to make a Nintendo game that actually works.”
 - “Our game is functional, so it actually works.”
- With appropriate scaffolds, **CT supported skill growth and a sense of belonging** among neurodiverse youth

Learners demonstrated agency through divergence and improvisation

- Many ignored or skimmed paper templates. They **chose to “just build.”**
- CT practices were **embodied in play**.
- **Abstraction, algorithmic thinking, and debugging** occurred in the Game Builder Garage engine.
- **Productive improvisation** included reframing bugs as features when this increased player enjoyment.
- **Peer help networks emerged:** learners demonstrated fixes, peer-taught game design structures, and co-debugged, suggesting redistributed expertise.

Discussion & Implications

Discussion:

- CAB feedback emphasized accessibility, inclusive recruitment, and educator readiness, which informed the initial G4G curriculum design.
- Pilot findings suggest that G4G is a promising tool that fostered computation thinking, creativity, and STEAM identity through hands-on, playful, and collaborative engagement.
- Learners demonstrated CT as an embodied and expressive practice by debugging, remixing, and solving problems through experimentation rather than using abstract worksheets.
- Findings reflect constructionist learning, where making, tinkering, and curiosity drive meaningful learning.

Curricular Insights & Refinement:

- The **CAB-informed design principles** (see Table 1) provided the foundation for refining the G4G curriculum.
- Pilot findings also unveiled important insights for refinement. Learners engaged most deeply when immersed directly in Game Builder Garage, preferring exploration over linear planning tools.
- Scaffolds and templates were often underused, increasing facilitator load but revealing opportunities for peer mentoring and shared problem-solving.

Future works

The next phase will scale pilot testing to larger and more diverse samples, track CT development over time, examine factors influencing sustained engagement, and refine interactive, multimodal scaffolds to support nonlinear, improvisational learning.

Table 1. Design principles drawn from CAB feedback

Design Principle	Descriptions
Accessibility First	Provide multiple modes of instruction (visual, audio, simplified guides) to reduce cognitive load.
Family Engagement	Involve parents to reduce stigma, build buy-in, and foster supportive home environments.
Inclusive Recruitment	Broaden outreach to diverse neurodiverse learners, ensuring equitable access and participation.
Evidence-Based Evaluation	Establish clear, multi-layered evaluation metrics (formative and summative) to measure participant growth in CT and STEAM identity.
Educator Preparedness	Provide training and adaptable resources so educators can confidently implement the G4G curriculum.

References

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